NGS Variant Analysis of SARS-CoV-2

Jânice Roberta de Paula

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Contents

1 Introduction **1**

2 Materials and Methods **2**

2.1 Reference Genome 2

2.2 Bioinformatics Tools 3

3 Results **3**

3.1 Sequencing Summary 3

3.2 Viral Genome Organization 4

3.3 Variant Detection 4

3.4 Mutation Distribution Across the Genome 5

3.5 Genome Coverage 7

4 Variant Quality Filtering **8**

5 Read Depth Analysis **11**

6 Allele Frequency Analysis **12**

7 Discussion **13**

8 Conclusion **14**

1 Introduction

The emergence of the novel coronavirus responsible for the global pandemic has highlighted the importance of genomic surveillance in understanding viral evolution and transmission dynamics. The virus known as SARS-CoV-2 is an RNA virus belonging to the Coronaviridae family and

possesses a single-stranded positive-sense RNA genome of approximately 29.9 kb.

Genomic sequencing has become a fundamental tool for tracking viral mutations, identifying emerging variants, and monitoring evolutionary changes that may affect viral transmissibility, pathogenicity, or immune escape. Advances in next-generation sequencing (NGS) technologies allow researchers to analyze viral genomes at high resolution and detect even low-frequency variants present in viral populations.

In this report, a genomic analysis of SARS-CoV-2 was performed using publicly available sequencing data obtained from the Sequence Read Archive (SRA). The sequencing reads were aligned to a reference genome, variants were identified, and mutation patterns across the viral genome were characterized.

The objectives of this analysis are:
Align sequencing reads to the reference genome
Identify genomic variants
Characterize mutations across the viral genome
Visualize genomic variation and sequencing coverage

This analysis demonstrates a typical bioinformatics workflow used in viral genomics and genomic epidemiology.

2 Materials and Methods

2.1 Reference Genome

The reference genome used in this analysis corresponds to the early SARS-CoV-2 isolate Wuhan-Hu-1 (accession number NC_045512.2). This sequence represents one of the first fully characterized genomes of the virus and has been widely adopted as the standard reference for SARS-CoV-2 genomic analyses.

The genome is approximately 29,903 nucleotides long and contains several open reading frames (ORFs) encoding both structural and non-structural proteins. Key structural genes include:

- Spike (S) – mediates viral entry into host cells
- Envelope (E) – involved in viral assembly
- Membrane (M) – the most abundant structural protein
- Nucleocapsid (N) – binds the viral RNA genome

The large ORF1ab region encodes the viral replication machinery, including RNA-dependent RNA polymerase and other enzymes required for viral replication.

2.2 Bioinformatics Tools

A standard viral genomics pipeline was implemented using widely adopted bioinformatics software tools.

The following tools were used:
BWA for read alignment: sequence alignment was performed using BWA, a fast and accurate aligner designed for mapping short sequencing reads against large reference genomes. This step identifies the genomic position from which each sequencing read originates.

Samtools for BAM processing: alignment processing and manipulation of sequencing files were conducted using SAMtools. This toolkit allows conversion between SAM and BAM formats, sorting of alignments, and indexing for efficient access.

Bcftools for variant calling: variant calling was performed using BCFtools, which analyzes the aligned reads to detect nucleotide differences between the sequenced sample and the reference genome. The output is stored in Variant Call Format (VCF), which includes information about variant position, quality scores, read depth, and allele frequency.

R for statistical analysis and visualization: statistical analysis, visualization, and report generation were performed in R (programming language) using graphical libraries for data visualization.

3 Results

3.1 Sequencing Summary

Table 1 summarizes the sequencing statistics obtained after alignment of reads to the SARS-CoV-2 reference genome.

These metrics provide an overview of sequencing quality and coverage depth across the viral genome.

Important metrics include:
- Total reads – the number of sequencing reads analyzed
- Mapped reads – reads successfully aligned to the reference genome
- Mapping rate – percentage of reads mapped to the genome
- Mean genome coverage – average sequencing depth across the genome
- Number of variants – mutations detected relative to the reference genome

High genome coverage is particularly important in viral sequencing because it increases confidence in variant detection and reduces the likelihood of sequencing artifacts.

Table 1: Sequencing summary statistics

Metric	Value
Total reads	436934
Mapped reads	0
Mapping rate (%)	0
Mean genome coverage	2034
Number of variants	21

3.2 Viral Genome Organization

The SARS-CoV-2 genome is organized into several coding regions that encode structural and non-structural proteins essential for viral replication and infection.

Table 2 lists the major genes and their genomic coordinates.

The ORF1ab region occupies approximately two-thirds of the genome and encodes non-structural proteins involved in viral replication and transcription. The remaining portion of the genome contains structural and accessory genes, including the Spike gene, which plays a key role in host cell entry through interaction with the ACE2 receptor.

Understanding the genomic organization of the virus is essential when interpreting mutation patterns, as mutations occurring in different genes may have distinct biological implications.

Table 2: Major genes in the SARS-CoV-2 genome

gene	start	end
ORF1ab	266	21555
Spike	21563	25384
ORF3a	25393	26220
E	26245	26472
M	26523	27191
N	28274	29533

3.3 Variant Detection

Variant calling analysis identified multiple nucleotide differences between the sequenced viral genome and the reference sequence.

Table 3 lists the detected variants along with several key metrics:

- Chromosome – reference genome identifier
- Position – genomic coordinate of the variant
- Reference allele – nucleotide present in the reference genome

- Alternative allele – nucleotide observed in the sample
- Quality score – statistical confidence of the variant call
- Depth – number of sequencing reads covering the position
- Allele frequency – proportion of reads supporting the variant

Variants with higher read depth and high allele frequency are generally considered more reliable indicators of true genetic variation.

The detected variants include both single nucleotide polymorphisms (SNPs) and small insertions or substitutions. Some mutations occur in regions encoding structural proteins such as the Spike gene, which may be relevant for viral fitness and immune escape.

Table 3: Detected genomic variants

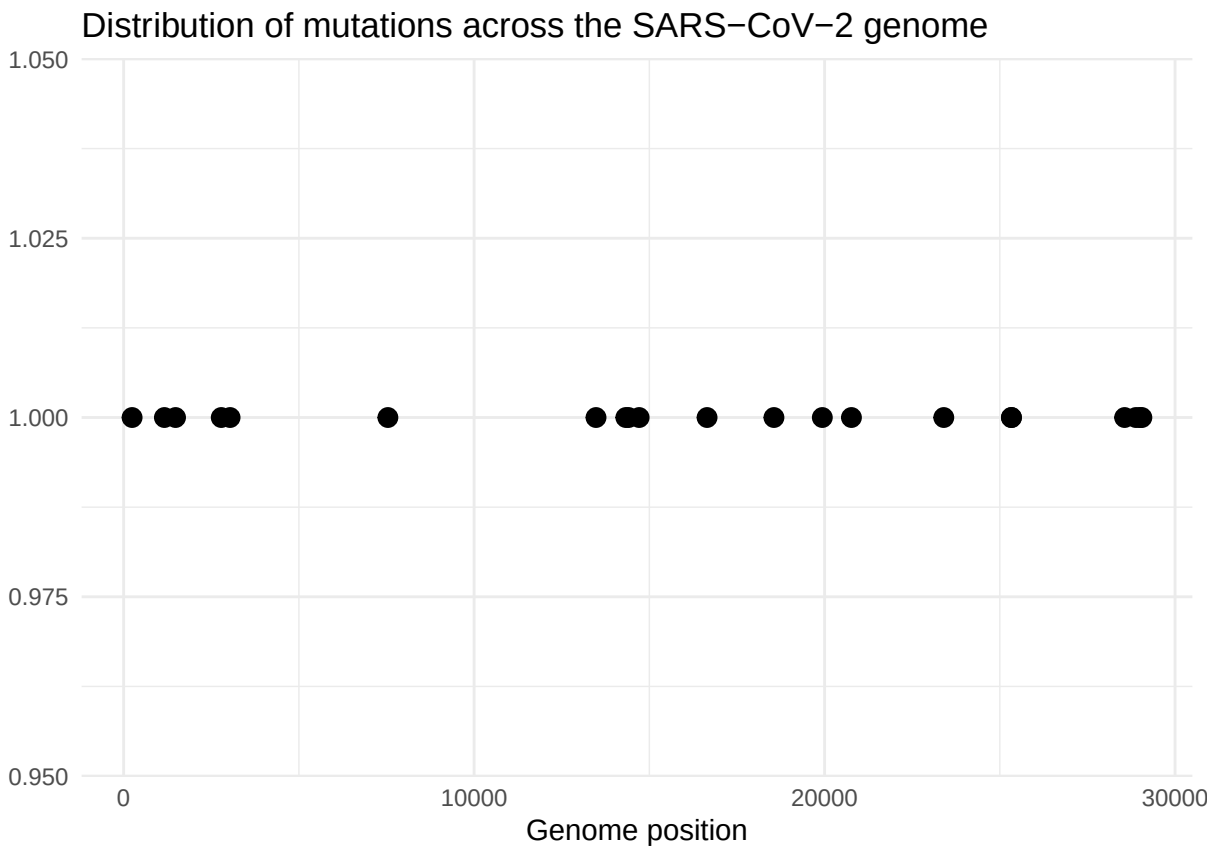
Chromosome	Position	Reference	Alternative	Quality	Depth	Allele_Frequency
NC_045512.2	241	C	T	186.9770	6	1.0
NC_045512.2	1163	A	T	21934.7000	684	1.0
NC_045512.2	1481	T	C	2783.5100	403	0.5
NC_045512.2	2785	T	C	0.0000	13	0.0
NC_045512.2	3037	C	T	27218.3000	853	1.0
NC_045512.2	7540	T	C	610.2410	20	1.0
NC_045512.2	13478	G	C	0.0000	33	0.0
NC_045512.2	14329	A	G	0.0000	26	0.0
NC_045512.2	14408	C	T	1012.8500	32	1.0
NC_045512.2	14709	TGT	TT	0.0098	14	0.5
NC_045512.2	16647	G	T	51910.6000	1598	1.0
NC_045512.2	18555	C	T	12848.9000	402	1.0
NC_045512.2	19935	T	C	0.0000	29	0.0
NC_045512.2	20766	A	T	0.0000	98	0.0
NC_045512.2	23401	GGA	AGG	386597.0000	12046	1.0
NC_045512.2	25330	T	A	0.0000	1536	0.0
NC_045512.2	25333	T	A	0.0000	1558	0.0
NC_045512.2	28561	T	G	0.0000	2524	0.0
NC_045512.2	28881	GGG	AAC	306.2830	11	1.0
NC_045512.2	28995	A	T	0.0000	24	0.0
NC_045512.2	29053	A	G	2222.3500	79	1.0

3.4 Mutation Distribution Across the Genome

To better understand the spatial distribution of mutations, the genomic positions of all detected variants were plotted across the SARS-CoV-2 genome.

This visualization reveals whether mutations are evenly distributed or concentrated in specific genomic regions. Clustering of mutations may indicate regions under selective pressure or areas with higher mutation rates.

In many RNA viruses, including SARS-CoV-2, certain mutation types occur more frequently due to the biochemical properties of the viral replication machinery and host-mediated RNA editing mechanisms.

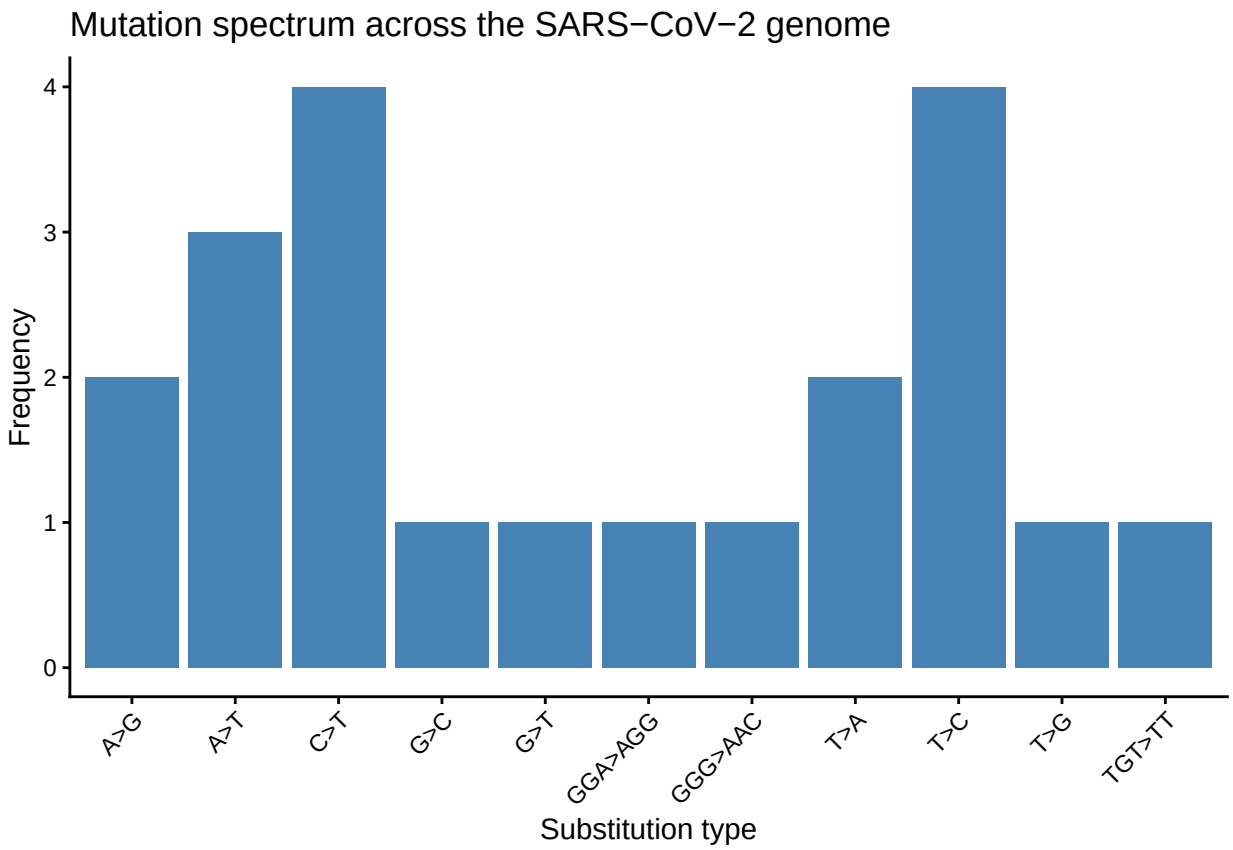


The mutation spectrum was also analyzed by classifying nucleotide substitutions according to their type (for example, C>T or A>G). Table 4 summarizes the frequency of each mutation type observed in the analyzed dataset.

Table 4: Mutation spectrum detected in the SARS-CoV-2 genome

Mutation_Type	Count
A>G	2
A>T	3
C>T	4
G>C	1
G>T	1
GGA>AGG	1
GGG>AAC	1
T>A	2
T>C	4
T>G	1

Mutation_Type	Count
TGT>TT	1



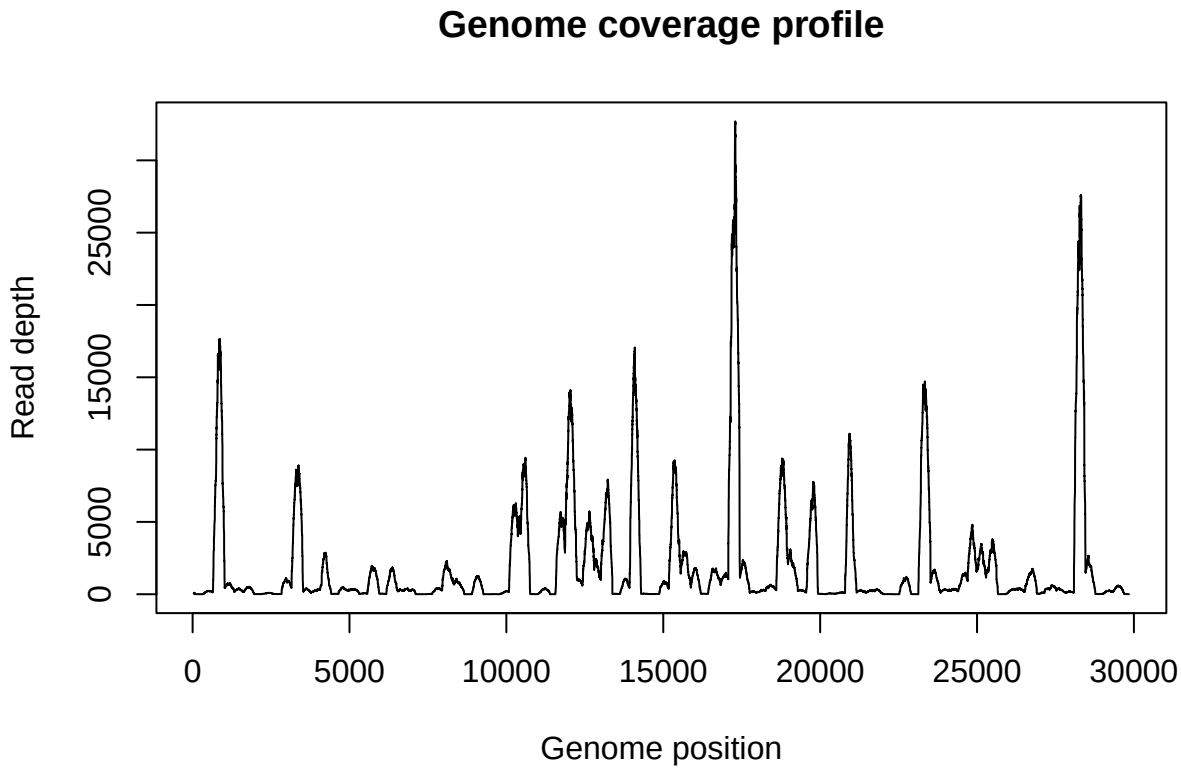
3.5 Genome Coverage

Genome coverage analysis evaluates the number of sequencing reads mapped to each position of the viral genome.

High coverage indicates that a genomic region has been sequenced multiple times, which increases confidence in variant detection. Conversely, regions with low coverage may be more prone to sequencing errors or missing variants.

The coverage profile plotted across the genome demonstrates that most genomic regions are covered by a substantial number of reads, indicating good sequencing depth and reliable variant detection.

Coverage analysis is a critical quality control step in genomic studies and helps ensure that variant calls are supported by sufficient sequencing evidence.



4 Variant Quality Filtering

Variant filtering was performed to remove low-confidence calls that could arise from sequencing errors or alignment artifacts. Only variants with a quality score greater than 30 were retained for downstream analysis. This threshold is commonly used in genomic studies to ensure reliable variant detection.

After filtering, the remaining variants were visualized across the viral genome to assess their genomic distribution and sequencing support.

Twenty-one variants were identified in the analyzed genome. After applying a quality filter (QUAL > 30), 11 variants were considered highly reliable, reducing the probability of false positives in the analysis of mutations present in the SARS-CoV-2 genome.

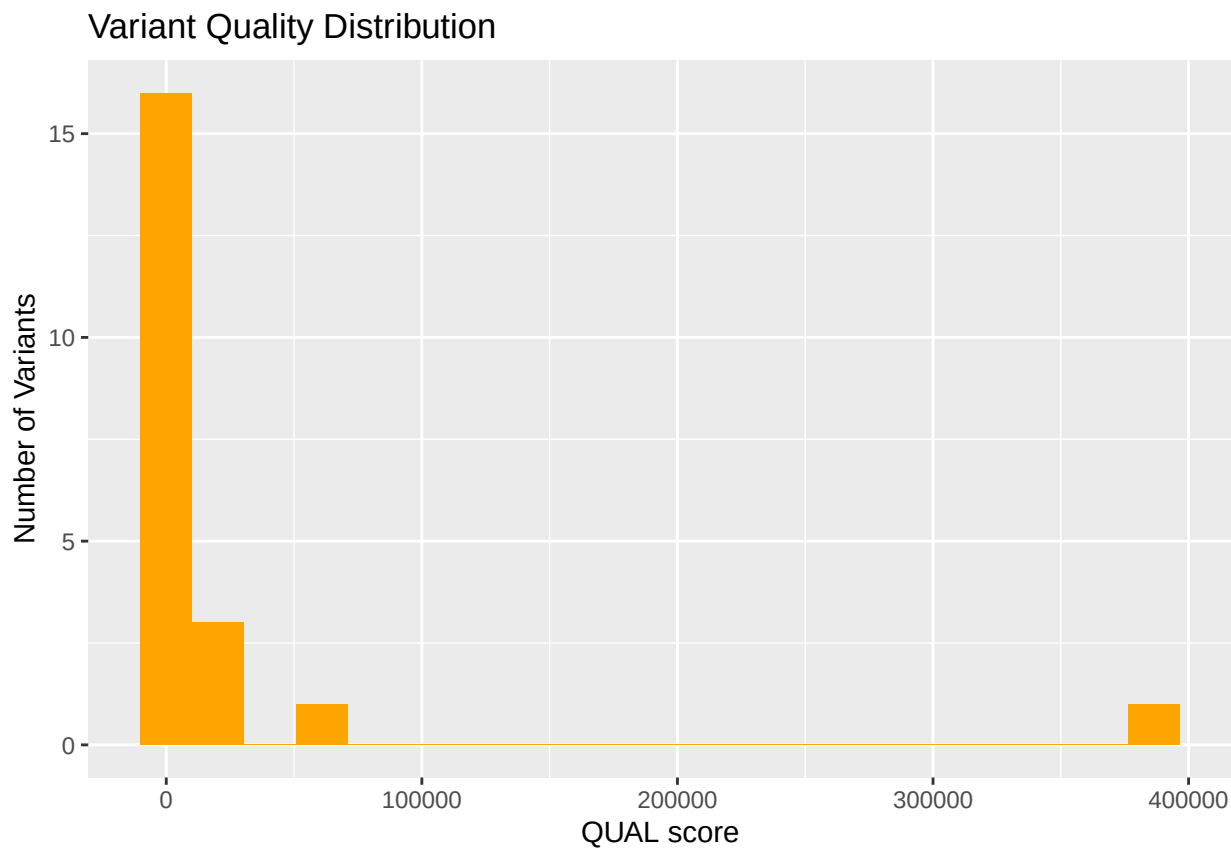
```
##                          seqnames start  end width strand paramRangeID REF ALT
## NC_045512.2:241_C/T  NC_045512.2   241  241     1      *         <NA>   C   T
## NC_045512.2:1163_A/T NC_045512.2  1163 1163     1      *         <NA>   A   T
## NC_045512.2:1481_T/C NC_045512.2  1481 1481     1      *         <NA>   T   C
## NC_045512.2:2785_T/C NC_045512.2  2785 2785     1      *         <NA>   T   C
## NC_045512.2:3037_C/T NC_045512.2  3037 3037     1      *         <NA>   C   T
```

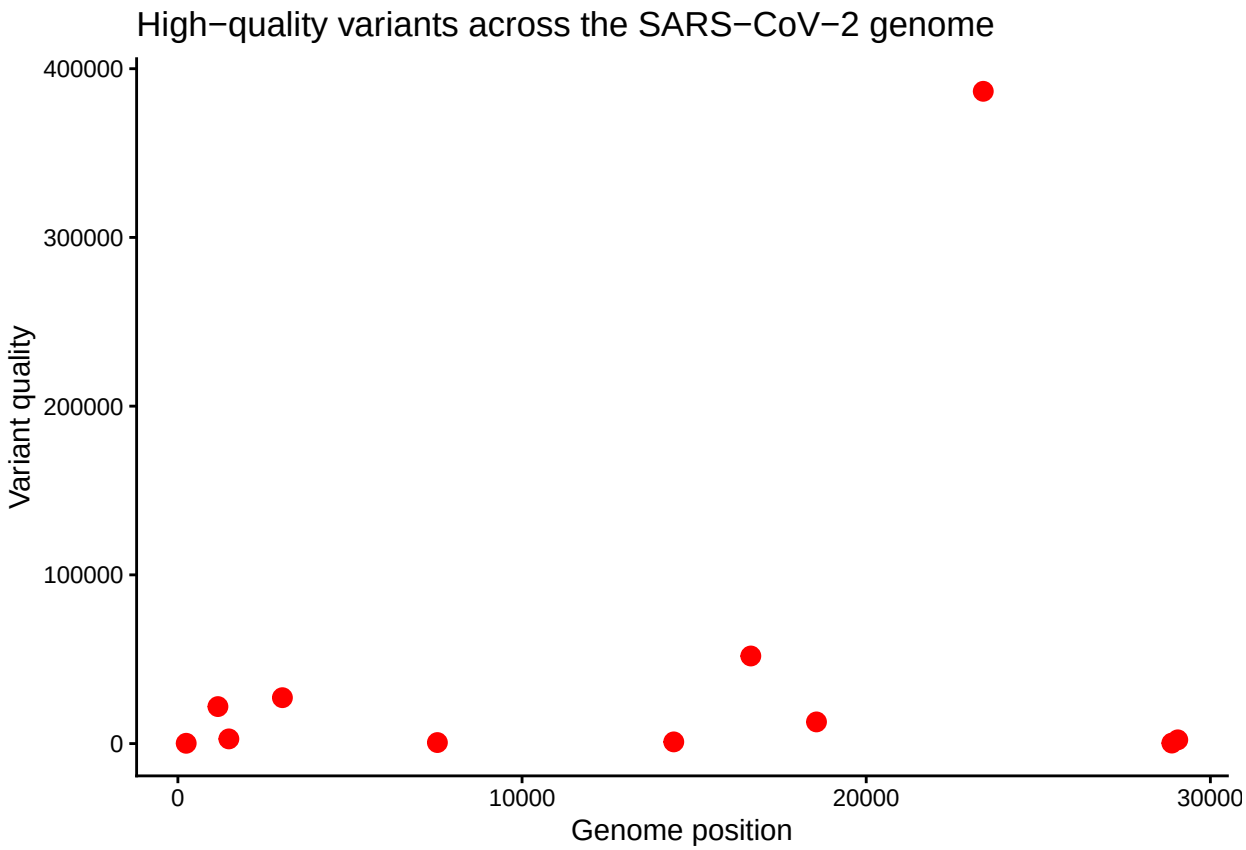
```
## NC_045512.2:7540_T/C NC_045512.2  7540 7540     1      *         <NA>   T   C
##                                 QUAL FILTER
## NC_045512.2:241_C/T    186.977000000      .
## NC_045512.2:1163_A/T 21934.700000000      .
## NC_045512.2:1481_T/C  2783.510000000      .
## NC_045512.2:2785_T/C     0.000008296      .
## NC_045512.2:3037_C/T 27218.300000000      .
## NC_045512.2:7540_T/C   610.241000000      .
```

```
## [1] 21
```

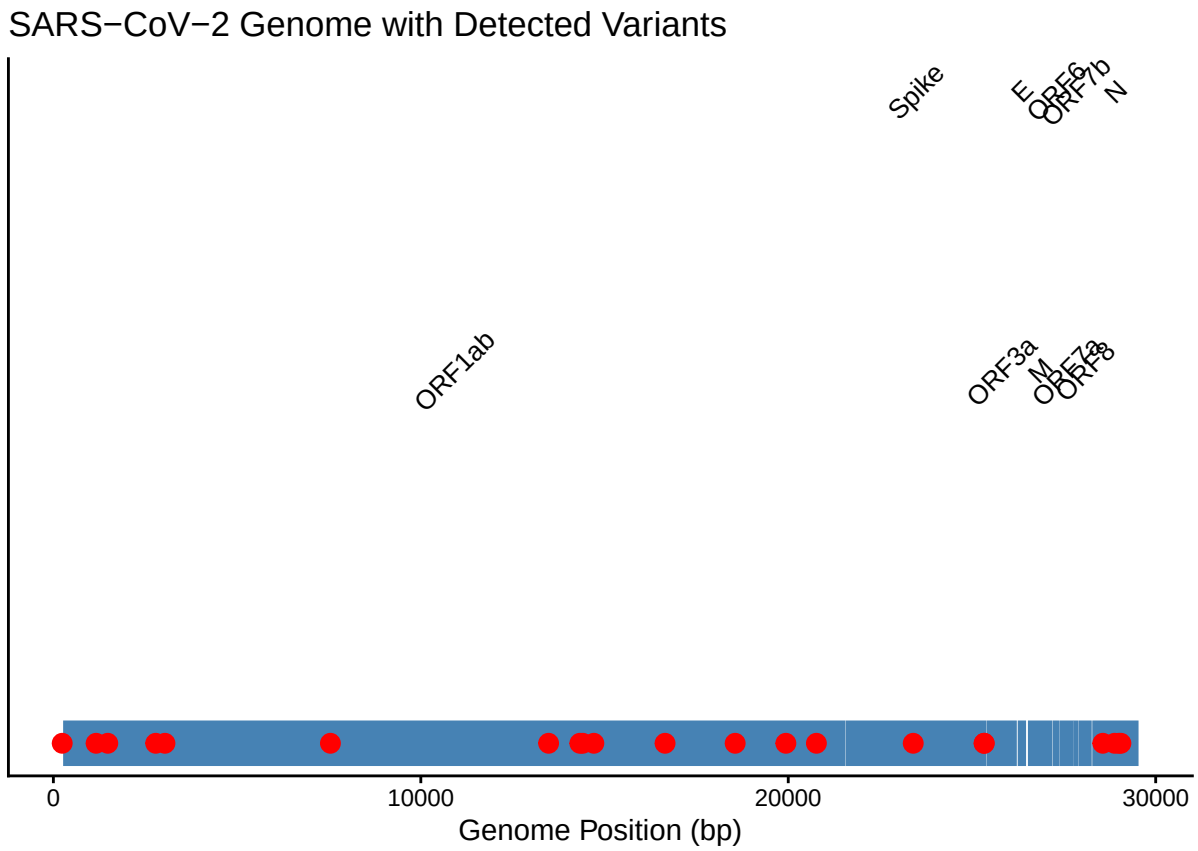
```
## [1] 11
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       0       0     187   24173    2784  386597
```






```
##         CHROM  POS REF ALT  DP  AF
## 1 NC_045512.2  241   C   T   6 1.0
## 2 NC_045512.2 1163   A   T 684 1.0
## 3 NC_045512.2 1481   T   C 403 0.5
## 4 NC_045512.2 2785   T   C  13 0.0
## 5 NC_045512.2 3037   C   T 853 1.0
## 6 NC_045512.2 7540   T   C  20 1.0
```

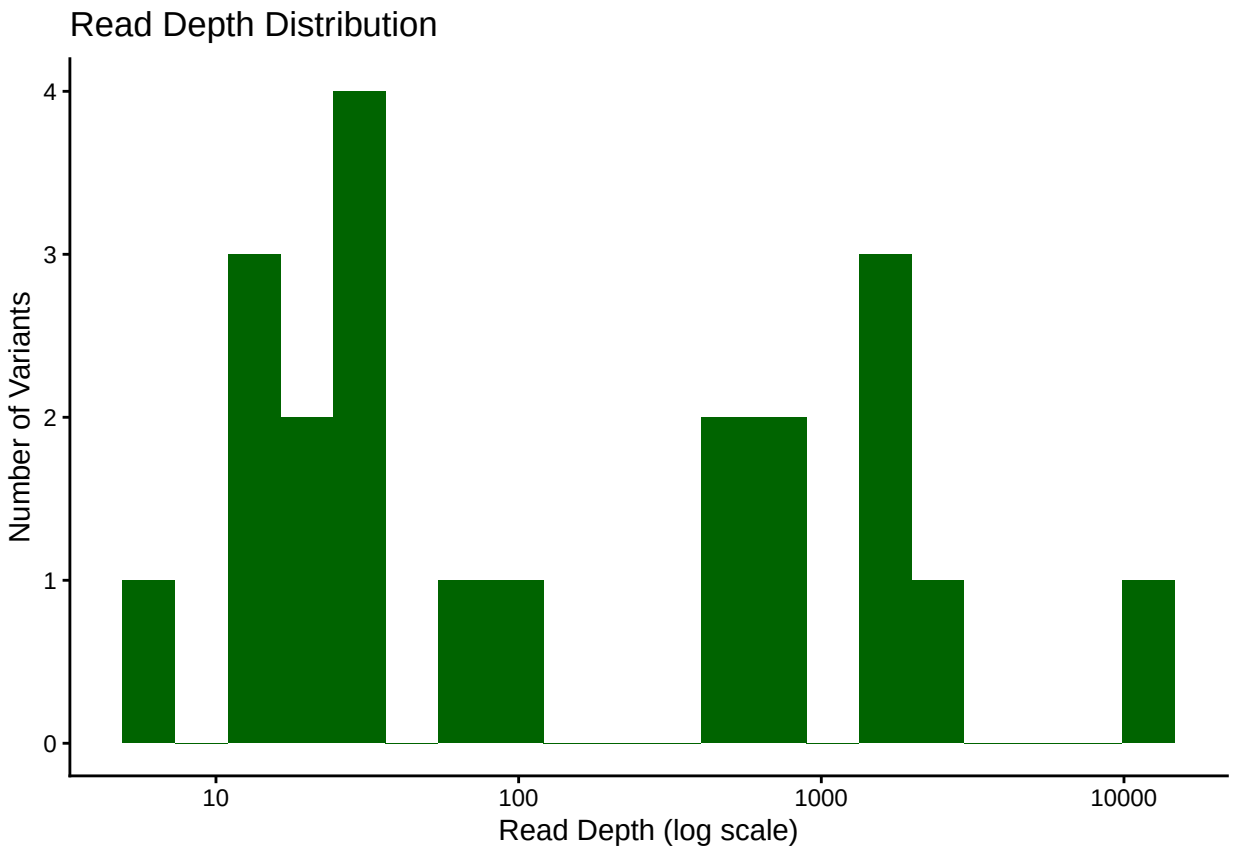


5 Read Depth Analysis

Read depth represents the number of sequencing reads covering a specific genomic position. Higher depth generally increases confidence in variant detection, as multiple independent reads support the observed nucleotide change.

The distribution of read depth across detected variants was analyzed to assess the reliability of variant calls across the genome of the SARS-CoV-2.

```
##         CHROM  POS REF ALT            QUAL
## 1 NC_045512.2  241   C   T   186.977000000
## 2 NC_045512.2 1163   A   T 21934.700000000
## 3 NC_045512.2 1481   T   C  2783.510000000
## 4 NC_045512.2 2785   T   C     0.000008296
## 5 NC_045512.2 3037   C   T 27218.300000000
## 6 NC_045512.2 7540   T   C   610.241000000
```

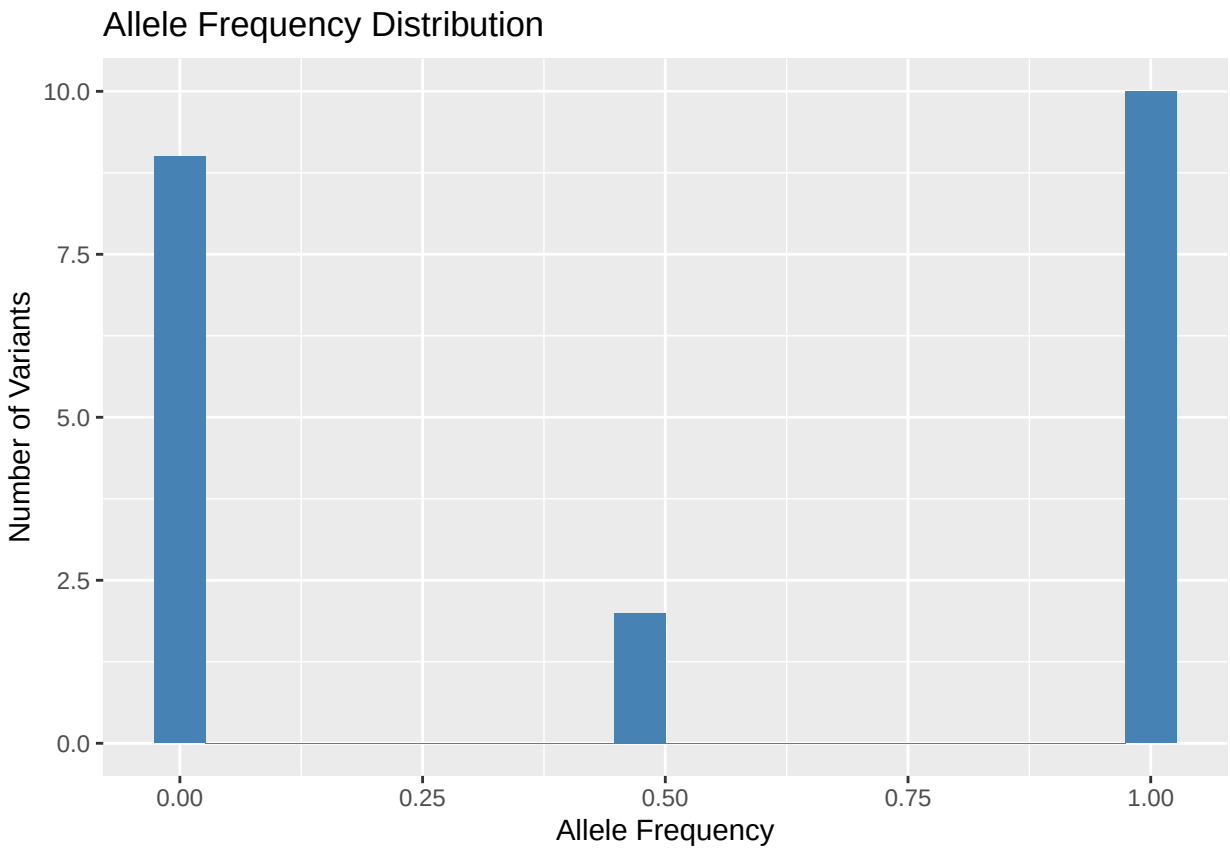


6 Allele Frequency Analysis

Allele frequency represents the proportion of sequencing reads supporting the alternative allele at a given genomic position. High allele frequencies suggest that the mutation is present in the majority of viral genomes in the sample, whereas low frequencies may indicate minor variants within the viral population.

The distribution of allele frequencies provides insights into the genetic diversity of the analyzed viral sample and helps distinguish dominant mutations from low-frequency variants in the genome of the SARS-CoV-2.

```
##                Variant Position  DP AC  AF  RO  AO
## 1  NC_045512.2:241_C/T      241   6  2 1.0   0   6
## 2 NC_045512.2:1163_A/T     1163 684  2 1.0   1 683
## 3 NC_045512.2:1481_T/C     1481 403  1 0.5 273 130
## 4 NC_045512.2:2785_T/C     2785  13  0 0.0  11   2
## 5 NC_045512.2:3037_C/T     3037 853  2 1.0   1 852
## 6 NC_045512.2:7540_T/C     7540  20  2 1.0   0  20
```



7 Discussion

The genomic analysis performed in this study identified multiple mutations across the analyzed SARS-CoV-2 genome. These mutations represent genetic differences between the sequenced viral isolate and the reference genome.

RNA viruses are characterized by relatively high mutation rates due to the error-prone nature of RNA-dependent RNA polymerase. As a result, viral populations often consist of a mixture of closely related genomic variants.

Mutations located in the Spike gene are of particular interest because this protein mediates viral entry into host cells and is the primary target of neutralizing antibodies. Changes in the Spike protein may influence viral infectivity, transmission efficiency, and immune escape.

Monitoring such mutations is therefore essential for understanding viral evolution and for guiding vaccine and therapeutic development.
The detected mutations reflect genetic variation in the analyzed SARS-CoV-2 isolate.
Mutations located in the Spike gene are of particular interest due to their role in viral entry and immune escape.

8 Conclusion

This study demonstrates a complete bioinformatics workflow for analyzing viral genomic sequencing data.

The analysis included:
- Alignment of sequencing reads to a reference genome
- Processing and filtering of alignment files
- Identification of genomic variants
- Visualization of mutation patterns across the viral genome
- Analysis of mutation spectra and genome coverage

Such analytical pipelines are widely used in viral genomics and genomic epidemiology to monitor emerging variants and track viral evolution.

The methods presented here illustrate how publicly available sequencing data can be used to investigate genomic variation in viruses and provide insights into viral genetic diversity.